Akwasi Tagoe

2156 N. Tull Ave. A3.
Fayetteville, AR 72704
linkedin.com/in/akwasi-tagoe

(479) 799-0090
atagoe@uark.edu

PROFESSIONAL SUMMARY

Ph.D. Candidate in Agricultural Education, Communications and Technology (Agricultural Systems Technology Management) with expertise in artificial intelligence (AI), machine learning (ML), computer vision, unmanned aerial systems (UAS), precision agriculture, and technology-enhanced agricultural education. Experienced in developing AI-driven learning, research, and decision-support systems that integrate UAV remote sensing, geospatial analytics, predictive modeling, and agricultural data science. Demonstrated success in teaching undergraduate and graduate students, mentoring emerging researchers, conducting interdisciplinary research, securing competitive funding, and translating advanced technologies into practical educational and agricultural applications. Committed to advancing the land-grant mission through innovative teaching, applied research, Extension programming, and stakeholder engagement.

EDUCATION

Ph.D., Agricultural Education, Communications & Technology Expected: Dec. 2026
Graduate Certificate in Entrepreneurship
University of Arkansas, Fayetteville, AR

M.Sc. Agricultural & Environmental Science Nov. 2016
Newcastle University, Newcastle Upon Tyne, United Kingdom

Postgraduate Certificate in Business Administration May 2014
Ghana Institute of Management & Public Administration, Accra, Ghana

B.Sc. (Hons) in Agriculture June 2010
Kwame Nkrumah University of Science & Technology, Kumasi, Ghana

RESEARCH INTERESTS

Artificial Intelligence and Machine Learning in Agricultural Education: AI-enhanced learning systems. Adaptive educational technologies. Intelligent tutoring and decision-support systems. AI literacy in agriculture.

Agricultural Technology Education: Digital agriculture education. Precision agriculture instruction. Technology adoption and workforce development. Experiential and technology-enhanced learning.

Precision Agriculture and Data Science: UAV-based phenotyping. Computer vision and image analytics. Geospatial data science. Predictive agricultural modeling.

Extension and Outreach Innovation: AI adoption in agriculture. Producer education and technology transfer. Workforce training in digital agriculture. Stakeholder engagement and capacity building.

TEACHING, MENTORING, AND EDUCATIONAL TECHNOLOGY

Teaching Assistant
University of Arkansas Jan. 2025 - Present

Unmanned Aerial Vehicles for Precision Agriculture (ASTM 5010V)

- Delivered lectures and laboratory instruction in UAV operations, flight planning, image acquisition, and agricultural remote sensing.
- Guided graduate students in applying geospatial technologies to agricultural problem solving.

Precision Agriculture Technology

- Assisted instruction on sensor technologies, agricultural automation, and data-driven decision making.
- Supported hands-on learning experiences integrating emerging agricultural technologies.

Sensors and Controls in Precision Agriculture

- Assisted with laboratory exercises focused on sensing systems, instrumentation, and precision agriculture applications.

Curriculum Development

- Re-developed laboratory exercises for Sensors and Controls course.
- Led transition of geospatial instruction from ArcMap to ArcGIS Pro.

Student Mentoring

- Mentored undergraduate researchers in: Artificial intelligence. Machine learning. UAV remote sensing. Python programming. Geospatial data science.
- Supported student participation in active research projects and conference presentations.

RESEARCH EXPERIENCE

Graduate Research Assistant Jan. 2024 - Present

Digital Agricultural Science and Technology Lab & Agricultural Robotics Initiative, University of Arkansas, Fayetteville, AR

Research focuses on integrating artificial intelligence, machine learning, remote sensing, and agricultural data systems to support precision agriculture, crop monitoring, and technology-enabled decision making.

Major accomplishments include:

- Developed AI-based computer vision systems for automated crop trait extraction.
- Implemented YOLO deep learning models for flower detection, canopy characterization, and vegetation assessment.
- Designed scalable geospatial workflows integrating UAV imagery, vegetation indices, environmental variables, and field observations.
- Conducted multidisciplinary research across horticultural, agronomic, and breeding systems.
- Integrated phenotypic and environmental datasets to support computational agriculture and data-driven crop improvement.
- Developed reproducible research pipelines using Python, R, GIS, and cloud computing environments.
- Mentored undergraduate researchers in AI and agricultural technology applications.
- Applied machine learning approaches including: Random Forest. Gradient Boosting. Regression Modeling. Predictive Analytics.

PROFESSIONAL EXPERIENCE

Cocoa Management Systems Coordinator Sept. 2019 – Jan. 2024

Ghana Cocoa Board (COCOBOD), Ghana

- Designed and implemented agricultural data management systems to standardize collection, validation, and reporting of production and yield data across 60+ cocoa-growing communities, supporting national-scale production planning and resource allocation.
- Developed and delivered training programs in agricultural monitoring and digital data collection workflows for 20+ field officers across multiple districts, improving reporting accuracy and reducing submission turnaround time.
- Led cross-functional data validation and quality assurance workflows with extension officers and partner organizations to reconcile field-level agricultural data with official production records.
- Developed scalable operational frameworks for agricultural monitoring systems, enabling expansion into additional cocoa-growing districts while maintaining data consistency and reporting integrity

Co-Founder & Agricultural Services Manager Oct. 2017 – May 2020

Nocofio FT Ltd., Ghana

- Co-founded an agricultural services enterprise providing agronomic advisory, input access, and farm monitoring support to 100+ smallholder farming households across multiple districts in Ghana.

- Designed scalable service delivery workflows integrating crop production planning, input logistics, and farm-level monitoring to support efficient agricultural advisory operations across geographically dispersed farming communities.
- Conducted farm assessments and developed data-informed agronomic recommendations that contributed to improved crop productivity and farm management outcomes among participating smallholders.

Agricultural Services Officer Jan. 2012 – Aug. 2017

Green Coast Foods, Ghana

- Implemented field monitoring programs across 50+ contracted farms to assess crop health, pest and disease incidence, and yield potential, improving supply forecasting and supporting commercial procurement planning
- Delivered field training and agronomic advisory support on crop production and post-harvest handling practices to improve product quality and compliance with commercial standards

PUBLICATIONS

Peer-Reviewed Journal Articles

- Yeboah, A., Tetteh, R., Mensah, E. O., Bonsu, B. B., Tagoe, A., & Boateng, I. D. (2025). Advances in crop germplasm conservation: A review of non-destructive techniques and their role in global food security. Journal of Agriculture and Food Research, 102435. https://doi.org/10.1016/j.jafr.2025.102435
- Tagoe, A., Silva, A., Koparan, C., Poncet, A., Wang, D., Johnson, D., & Worthington, M. (2024). Blackberry growth monitoring and feature quantification with unmanned aerial vehicle (UAV) remote sensing. AgriEngineering, 6(4), 4549. https://doi.org/10.3390/agriengineering6040260

Journal Articles Under Review

- Tagoe, A., Koparan, A., Worthington, M., Johnson, D., Wang, D., & Wardlow, G.W. (2026). Object Detection Algorithm Evaluation on Blackberry Flower Quantification from UAV Imagery. Journal of Agriculture and Food Research. (Under review).
- Tagoe, A., Azmir, N.M., Nores, M., Elli, F.E., Johnson, D., Wardlow, G.W., & Koparan, A. (2026 UAV-Assisted Corn Biomass Estimation from RGB Monocular Depth). Journal of Agriculture and Food Research. (Under review).

Referred Conference Proceedings

- Tagoe, A., Bist, R. B., Koparan, C., Johnson, D., Wang, D., Worthington, M., & Poncet, A. (2025). Preliminary analysis of computer vision for blackberry flower quantification. Proceedings of the 2025 ASABE Annual International Meeting. doi:10.13031/aim.202501350
- Azmir, M. N., Tagoe, A., Koparan, C., Burgos, N. R., Davis, J., Runkle, B., & Wang, D. (2025). Automated weed pressure measurement system evaluation for unmanned aerial vehicles. Proceedings of the 2025 ASABE Annual International Meeting. doi:10.13031/aim.202501523
- Tagoe, A., Koparan, A., Koparan, C., Poncet, A., Wang, D., Johnson, D. M., & Worthington, M. (2024). Vegetation-coverage-specific flower density estimation in blackberry using unmanned aerial vehicle (UAV) remote sensing. 16th International Conference on Precision Agriculture. International Society of Precision Agriculture.

PRESENTATIONS AND RESEARCH DISSEMINATION

National and International Venues

- Tagoe, A. (2026). UAV-Based Blackberry Yield Prediction with Feature Quantification and Statistical Learning. Oral presentation, AI in Agriculture and Natural Resources Conference, Raleigh, MS. April 2, 2026.
- Tagoe, A. (2025). Preliminary analysis of computer vision for blackberry flower quantification. Oral presentation, American Society for Horticultural Science, New Orleans, LA. July 30, 2025.
- Tagoe, A. (2025). Preliminary analysis of computer vision for blackberry flower quantification. Poster presentation, ASABE Annual International Meeting, Toronto, Canada. July 15, 2025.
- Tagoe, A. (2025). Preliminary analysis of computer vision for blackberry flower quantification. Poster presentation, AI in Agriculture and Natural Resources Conference, Starkville, MS. April 1, 2025.

- Tagoe, A. (2024). Vegetation Coverage Specific Flower Density Estimation in Blackberry Using UAV Remote Sensing. Poster presentation, International Conference of Precision Agriculture (ICPA), Manhattan, KS. July 23, 2024.
- Tagoe, A. (2024). Vegetation Coverage Specific Flower Density Estimation in Blackberry Using UAV Remote Sensing. Poster presentation, S1069 Multistate Research Meeting, Montana State University, Bozeman, MT. June 13, 2024.

University of Arkansas

- Tagoe, A. (2025). Deep Learning Approach for Blackberry Feature Quantification in Plot-Level UAV Imagery. Poster presentation, Dale Bumpers Ag. Poster Event, University of Arkansas. October 30, 2025.
- Tagoe, A. (2025). Deep Learning Approach for Blackberry Feature Quantification in Plot-Level UAV Imagery. Poster presentation, ASABE Arkansas State Sectional Meeting, Little Rock, AR. October 3, 2025.
- Tagoe, A. (2024). Drones, Data, Blackberries. Finalist presentation, 3 Minutes Thesis (3MT) Competition, University of Arkansas, Fayetteville, AR. November 22, 2024.
- Tagoe, A. (2024). Vegetation Coverage Specific Flower Density Estimation in Blackberry Using UAV Remote Sensing. Poster presentation, ASABE Arkansas State Sectional Meeting, University of Arkansas. October 4, 2024.

Outreach, Capacity Building, and Stakeholder Engagement

- Tagoe, A. (2026). Unmanned Aerial Vehicles for Precision Agriculture. Invited presentation to Green Forest High School Students, Abernathy Agriscience & Technology Education Center (ASTC), Fayetteville, AR. January 14, 2026.
- Tagoe, A. (2024). Unmanned Aerial Vehicles for Phenotyping. Invited presentation to JR MANRRS Scholars (6th-11th grade), Abernathy AgriScience & Technology Education Center (ASTC), Fayetteville, AR. July 18, 2024.

LEADERSHIP AND SERVICE

- GPSC At-Large Representative, Dale Bumpers College of Agricultural, Food and Life Sciences, University of Arkansas (2025) - represented graduate student interests in college-level policy and funding decisions
- Member, ASABE MS-60 Unmanned Aerial Systems Technical Committee (2025) - contributed to technical standards development for UAS applications in agriculture
- Coordinator, UARK-Ghana Students Association (2025)

HONORS, AWARDS, AND COMPETITIVE FUNDING

- Kaneaster & Ruth Lindley Hodges Entrepreneurial Fellowship Award - $10,000, University of Arkansas (May 2025)
- GPSC Tiffany Marcantonio Research Grant - $1,500, University of Arkansas (April 2025)
- National Grain and Feed Association (NGFA) Scholarship - $5,000, University of Arkansas (December 2024)
- USDA-NIFA Agricultural Genome to Phenome Initiative Travel Grant - $1,500 (June 2024)
- S1069 Research and Extension for UAS Applications in U.S. Agriculture and Natural Resources Travel Grant - $1,500 (June 2024)
- 3 Minute Thesis (3MT) Finalist - competed among top doctoral researchers'university-wide, University of Arkansas (2024)

PROFESSIONAL AFFILIATIONS AND CERTIFICATIONS

- FAA Part 107 Remote Pilot Certificate for Small Unmanned Aircraft Systems
- American Society of Agricultural and Biological Engineers (ASABE)
- American Society for Horticultural Science (ASHS)
- International Society of Precision Agriculture (ISPA)
- Minorities in Agriculture, Natural Resources and Related Sciences (MANRRS)